

Design and Implementation of a Hierarchical Diagnosis-Assisted Tool on HarmonyOS

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Abstract—This study describes the design and development of an intelligent medical app that combines the DeepSeek-R1 model with HarmonyOS technology. The app offers key features like smart symptom search, online consultation, and diagnostic assistance. After introducing the core technologies, the system's overall architecture and module functions were outlined, and the main components were implemented through coding. The goal of this system is to improve healthcare delivery, make smart medical tools more accessible, and support the digital transformation of the healthcare industry.

Keywords—*hierarchical diagnosis and treatment, DeepSeek-R1, HarmonyOS, symptom retrieval*

I. INTRODUCTION

In recent years, many countries and regions around the world have been facing common challenges such as uneven allocation of medical resources and weak primary healthcare service systems. Data show that in most developing countries, and in some developed countries, patients generally prefer to seek care at large general hospitals rather than primary care facilities[1-3]. This phenomenon leads to the overconcentration of high-quality medical resources in major hospitals, underutilization of community-level services, and subsequent issues such as reduced system efficiency, prolonged wait times, and increasing tension between healthcare supply and demand, commonly referred to as "medical congestion"[4].

In order to alleviate this problem, China has actively promoted the establishment of a medical service model centered on "hierarchical diagnosis and treatment and two-way referral", and has gradually promoted the realization of a structured referral system featuring primary care, emergency and chronic care, up-and-down linkage, and information sharing[5].

Different countries have developed structured healthcare models that aim at enhancing resource allocation and improving healthcare quality. The UK's "gatekeeper system", based on family doctors, requires patients to see a GP first, who then decides if they need specialist care[6]. Likewise, Japan's "three-tiered healthcare system" is structured into three distinct levels—primary, secondary, and tertiary care—to ensure a more balanced and effective distribution of healthcare services across different tiers. However, due to multiple factors such as system design, technical support, and patient awareness, the implementation of such models still faces many challenges on a global scale[7].

At the same time, the rapid development of artificial intelligence models and other cutting-edge technologies has brought about brand new opportunities in the medical field, showing great potential in terms of AI-assisted diagnosis and treatment, medical image analysis, disease prediction, and personalized medicine[8]. At present, the emergence of big models such as Deepseek-R1 accelerates the development of medical AI and promotes the change of data-driven medical models. By integrating multimodal medical data and deep pre-training, such models significantly enhance their application value in decision support, accurate diagnosis, and clinical optimization[9].

Based on the above background, we mainly discuss how to effectively integrate the concept of hierarchical diagnosis and treatment into the medical process with the help of large-scale AI model technology, and apply it to the development of the "Smart Healthcare Guardian" software. Patients can use voice or text to describe their conditions to the system, and the platform, relying on the built-in database, deeply interacts with the big model to analyze the patient's condition and match relevant medical information to the patient. The system is also able to push more suitable medical institutions for the patient based on the patient's health condition and medical needs, realizing hierarchical diagnosis and treatment, and improving the efficiency and accuracy of the patient's access to medical resources.

II. RELATED TECHNOLOGIES OF THE SYSTEM

A. DeepSeek-R1

DeepSeek-R1 is an advanced AI language model introduced by the DeepSeek team, combining Mixture of Experts (MoE) to improve computational efficiency, Group Relative Policy Optimization (GRPO) to strengthen the optimization decision-making capability of learning, and Multi-Head Latent Attention (MLA) to improve the decision-making capability of processing long text. GRPO enhances the decision-making ability of optimization learning, MLA enhances the processing of long text, and Multi-Token Prediction (MTP) improves the coherence of text generation, especially in Chinese Natural Language Processing (NLP) tasks. Meanwhile, the model provides a lightweight version to support mobile operation, and the model is based on the MIT open source protocol, which facilitates secondary development and academic research. Although this openness may place some burden on users who need to manage model updates or adapt the framework to different computing environments[10].

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B. HarmonyOS

The HarmonyOS system is the “brain” that manages computer hardware resources and is an important carrier for allocating hardware resources and implementing application software functions, enabling the cooperative connection of multiple devices and providing strong security capabilities. This system is designed and developed based on the HarmonyOS platform, with the help of its distributed ability to achieve data synchronization, to ensure that the medical consultation function can be adapted to cell phones, tablets, smart screens, and other devices. The characteristics of HarmonyOS’s lightweight kernel ensure that the system runs smoothly on low-power devices[11].

C. ArkUI Front-End Framework

ArkUI is a declarative UI framework provided by HarmonyOS, which supports efficient rendering and cross-device adaptation. This system adopts ArkUI’s Column, Row, List, and other components for page layout, and uses its Swiper component to realize the rotating display of medical information. ArkUI’s state management mechanism can make the page dynamically update the UI. In the process of message interaction, the front-end can automatically render the dialog content through the received server-side data.

III. OVERALL SYSTEM DESIGN

A. System functional structure design

For the functions of the Smart Healthcare Guardian system, it is divided into several modules: intelligent condition retrieval, online diagnosis and treatment, appointment registration, knowledge popularization, service evaluation, etc. Together, these modules constitute the overall framework of the system, which can play a favorable support role for the Smart Healthcare APP. The functional structure diagram of the Smart Healthcare Guardian system is shown in Figure 1, which clearly demonstrates the hierarchical relationship between the system objectives and each specific function.

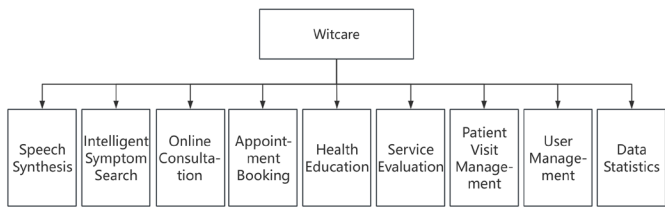


Fig. 1. Functional Architecture of the System

B. System Process Design

The Smart Healthcare Guardian system operates through two core elements: a patient-accessible frontend interface and an administrative backend unit. The frontend delivers medical support features—including intelligent symptom retrieval, appointment scheduling, online diagnosis, and inquiry publishing—whereas the backend conducts automated data management and content validation, covering identity authentication, encrypted consultation archives, diagnostic result certification, and evidence-based medical knowledge updates.

The core business data flow diagram (shown in Figure 2) describes in detail the entire process—from the user’s input of

medical condition description to the acquisition of medical services—as well as the processing and auditing of data by the backend management system.

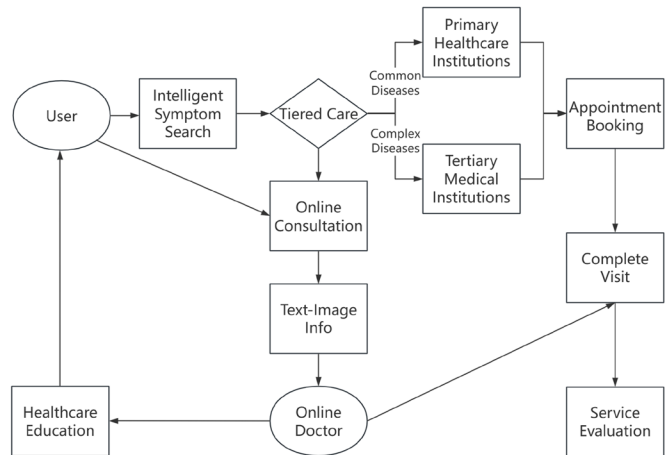


Fig. 2. Core Business Data Flow Diagram of the System

C. System Architecture Design

In order to better meet user needs, the architecture design of the Smart Healthcare Guardian system is divided into five layers: access layer, performance layer, application layer, support layer, and foundation layer. Each layer is logically independent of the other, focusing on modularization, with high cohesion and low coupling characteristics. At the same time, the architecture design aims at guaranteeing the security, reliability, simplicity, and scalability of the system. The overall system architecture is shown in Figure 3.

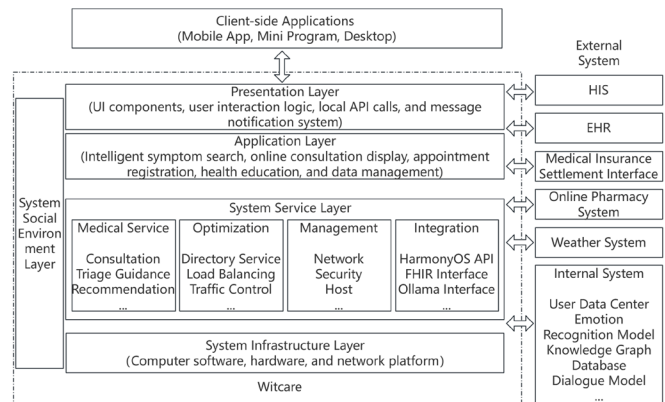


Fig. 3. Overall System Architecture

D. System Database Design

The Smart Healthcare Guardian system uses MySQL with a relational schema tailored to its specific functions. It includes multiple tables to organize various types of data. Key tables include Diseases, Users, Diagnosis, and Finances, each responsible for storing distinct categories of information.

Diseases Table: Stores information related to the disease database, including fields such as disease ID (_id), disease name (diseaseName), symptoms (symptoms), treatment recommendations (recommendation), corresponding medical institution (institution), and disease category (category).

Users Table: Stores basic user information, including the user's unique identifier (`_openid`), username (`name`), password (`pwd`), and avatar file path (`avatarUrl`).

Diagnosis Table: Records patient consultation details on the platform, such as diagnosis start time (`start_Dtime`), patient identifier, and physician identifier.

Finances Table: Logs patient financial records, including fields like consumption time (`consumption_time`), transaction type (`fin_type`), amount (`money`), and patient identifier.

Table 1 presents the detailed schema of the Diseases table as an example.

TABLE I. SCHEMA OF THE DISEASES TABLE

Field Name	Data Type	Constraint	Description
<code>_id</code>	<code>bigint(20)</code>	Not NULL	Disease ID, primary key
<code>diseaseName</code>	<code>varchar(100)</code>	Not NULL	Name of the disease
<code>symptoms</code>	<code>varchar(255)</code>	Not NULL	Common symptoms
<code>recommendation</code>	<code>varchar(255)</code>	Not NULL	Diagnostic and treatment recommendations
<code>institution</code>	<code>varchar(100)</code>	Not NULL	Recommended medical institution
<code>category</code>	<code>varchar(50)</code>	Not NULL	Disease classification
<code>prevention</code>	<code>varchar(255)</code>	Not NULL	Preventive measures
<code>diNote</code>	<code>varchar(255)</code>	NULL	Additional notes

IV. SYSTEM FUNCTION IMPLEMENTATION

A. Login Interface

The system utilizes the HarmonyOS account framework for user authentication. Users can log in through mobile phone number, HarmonyOS account, or third-party authorization such as WeChat. Upon the user's first login, the system invokes the Account Kit API to obtain user authorization information and stores the user's basic profile data in the Users table of the database.

B. Main Interface

The main interface focuses on the text-and-image consultation feature and uses ArkUI's Column and Row layout components. It has two sections: the carousel screen at the top and the scrolling panel under it. The carousel screen uses a Swiper component inside a List component to show rotating medical information and health news to users. Below the carousel screen, the scrolling panel uses a Column layout to show more consultation content like health education, disease knowledge, and related information in list or card formats.

The main interface provides a functional option area using a Grid layout, providing users with shortcut entrances to the main interface, online consultation, personal services, etc. It supports users in triggering event processing functions through interactive buttons to send requests to the back-end services. The back-end adopts RESTful API architecture. The Controller layer receives user requests to call the Service layer to access the database, retrieve the matching consultation content through the medical consultation table, and return to the front-end to realize the dynamic update of medical information. The main interface is shown in Figure 4.

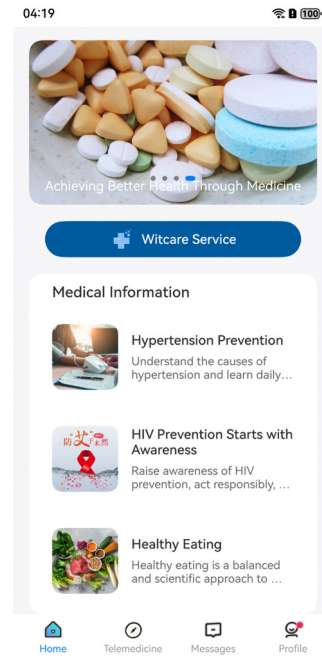


Fig. 4. Main Interface

C. Personal Interface

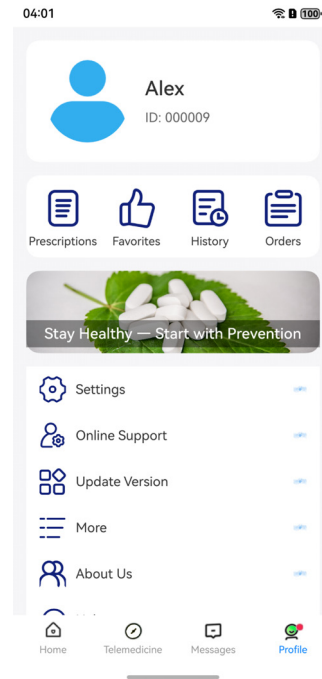


Fig. 5. Personal Interface

The personal interface provides users with personalized information display pages, such as user avatars, nicknames, graphic collections, and following doctors, and adds incentives such as doctor cards and coupons to increase user stickiness and activity. In addition, users can also query the history of medical treatment, view e-prescriptions issued by online doctors, and obtain health intervention suggestions pushed by the system.

through the personal interface. The personal interface is shown in Figure 5.

Architecturally, the module mainly uses ArkUI's Form and List components to realize information display, and is associated with the cloud database to ensure that users' personal information and medical history data can be accessed and updated at any time.

D. Community Messaging Module

The Community Messaging module provides functions such as article comments, video communication, patient interaction, etc. It also integrates online medical consulting services to enable efficient communication between patients and doctors. At the same time, users can also create disease group chats to provide social communication for users. For example, users with rare diseases can share their treatment experiences and encourage each other in specific groups; parents concerned about the education of autistic children can also share their experiences with each other in specialized groups. The messaging interface is shown in Figure 6.

The system adopts ArkUI's List and Chat components to build the instant messaging interface, and realizes real-time pushing of chat messages by using the WebSocket protocol, and all users' interactive messages are stored in the Messages table of the cloud database, which ensures the persistent storage of conversation records.

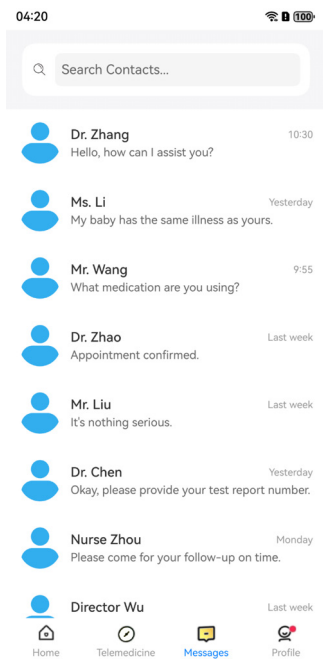


Fig. 6. Messaging Interface

E. Intelligent Symptom Retrieval Module

The Intelligent Symptom Retrieval module supports the user in inputting symptom information and provides possible disease suggestions and consultation recommendations based on the deployed R1 model for preliminary analysis. The user can enter text to describe the condition through the TextInput component of ArkUI or the speech-to-text function.

After receiving the user's complaint information, the system calls the server and combines it with the professional medical corpus to intelligently determine the possible causes of the disease and provide the corresponding consultation recommendations, recommending patients who tend to suffer from common diseases and multiple diseases to the primary healthcare institutions, and patients who tend to suffer from critical and difficult diseases to large hospitals, so as to increase the utilization rate of healthcare resources.

In Figure 7, it is shown that when a user experiences abdominal discomfort, the system provides the user with a series of possible disease suggestions based on the available information. When the user provides information, the system will reason about various conditions, dynamically speculate on the potential cause of the condition, and give the user the corresponding diagnosis and treatment suggestions, instead of directly giving an exact diagnosis. This is because the development of the disease is uncertain, and the system is intended to provide the user with a reference rather than a final diagnosis, so the user needs to decide on their own based on the probability situation.

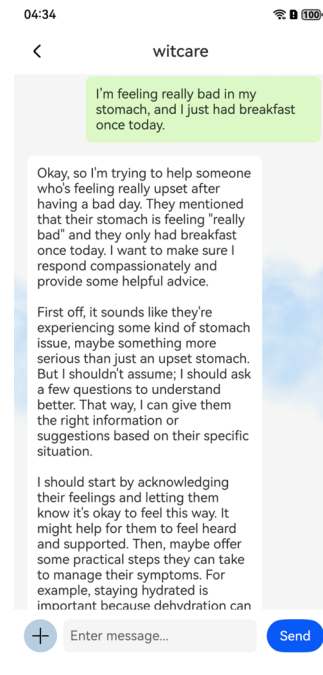


Fig. 7. Intelligent Symptom Retrieval Interface

When users provide more details, the system gradually gets better at understanding their symptoms and focuses its search to offer clearer advice. If medical help is needed, it determines how serious the condition is and recommends the most suitable healthcare facility. In these situations, guiding patients to nearby community hospitals work best.

F. Online Diagnosis And Treatment Module

The online diagnosis and treatment part module telemedicine, together with HarmonyOS cloud services to offer consultations in real time. Information about doctors is kept in a Doctors database, which helps organize and recommend doctors so users can quickly find the right one. After confirming their

condition, users pick a recommended doctor to begin an online consultation. The online consultation interface is shown in Figure 8.

During the consultation, users can type messages using ArkUI's TextInput, upload photos or videos, and take part in voice or video calls. Doctors use video calls to communicate with patients more directly and clearly.

After diagnosis, doctors can send electronic prescriptions via the system directly to users. Patients can buy medicines online or collect them at physical pharmacies. When the doctor considers the case serious, they may suggest that the patient visit a hospital in person for further care.

Before selecting a doctor for online consultation, the system will provide real-time updates of the doctor's consultation status to the user, displaying the doctor's current consultation status and estimated waiting time. Based on this information, the user can choose whether or not to book an appointment with the current doctor. Each doctor-patient consultation record will be stored in the system's database, which is convenient for users to check the history of consultation records at any time. Users who are not satisfied with the doctor's service can also apply for a refund or change the doctor through the customer service module.

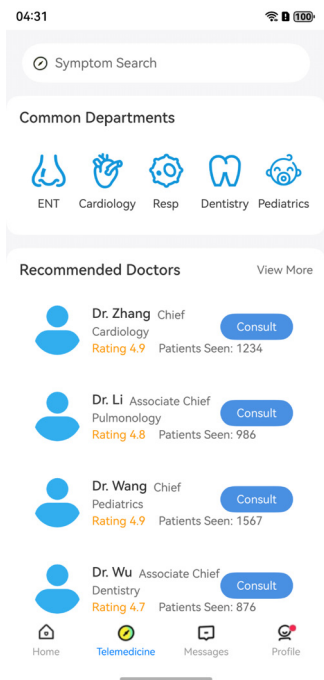


Fig. 8. Online Consultation Interface

V. CONCLUSION

A smart healthcare APP based on a big model and HarmonyOS was designed and developed, which includes

intelligent symptom retrieval, online consultation, and assisted diagnosis and treatment. The system is able to call DeepSeek-R1 to conduct a preliminary analysis of the condition information entered by the patient through multimodal medical data, and provide users with efficient and convenient medical consulting services. This has the advantage of high efficiency compared to the traditional medical model, and can reasonably optimize the distribution of medical resources while promoting the long-term development of China's medical career.

Although the software has entered the development stage, there are still some difficulties in the construction of the condition database. The diagnosis and treatment recommendations generated by the AI model have a certain reference value, but they cannot be used as a direct basis, and there are still some errors. These results need to be judged in combination with the knowledge of clinical medicine. In the future, we will continue to study these problems and solve them to further improve our project.

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